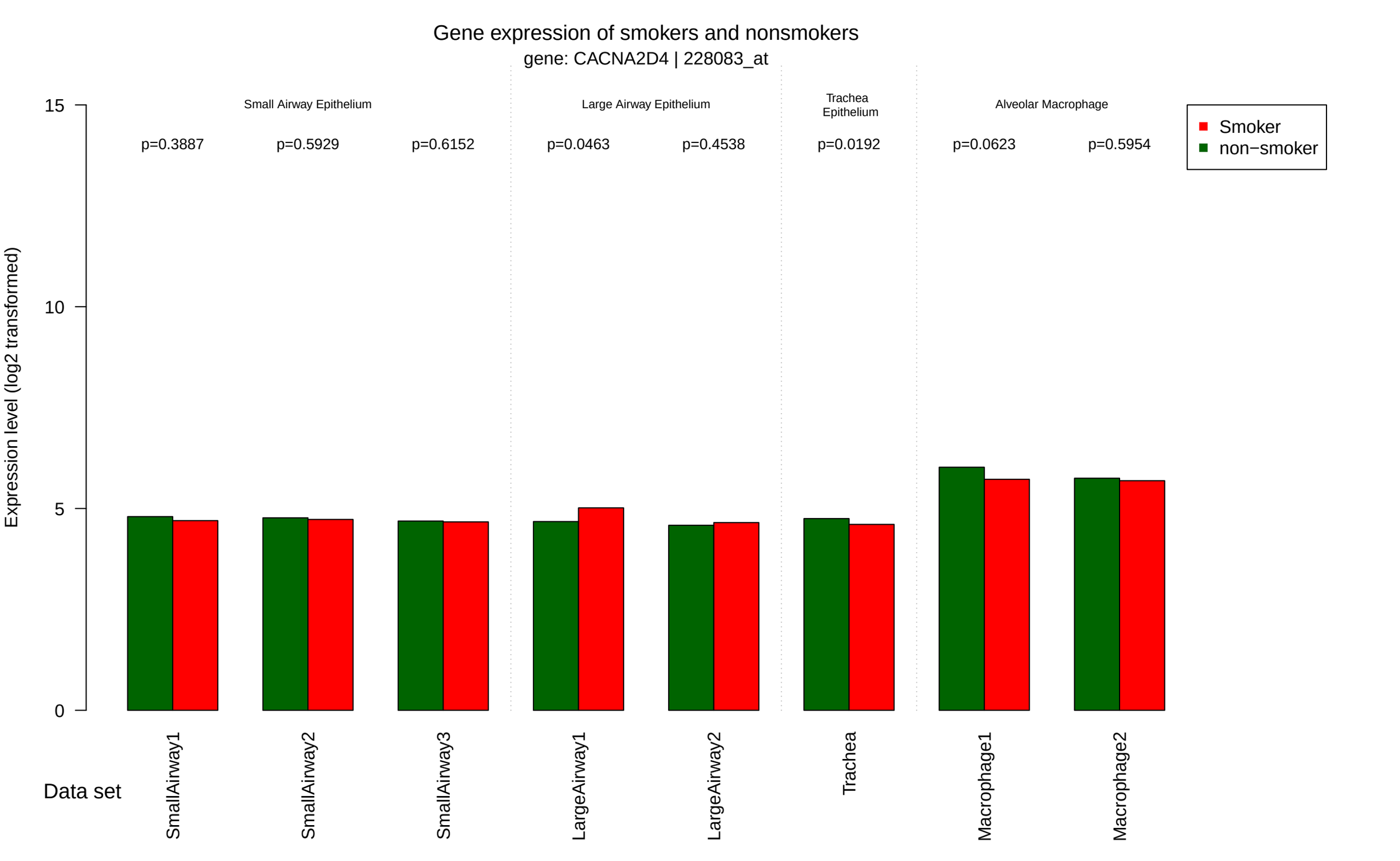
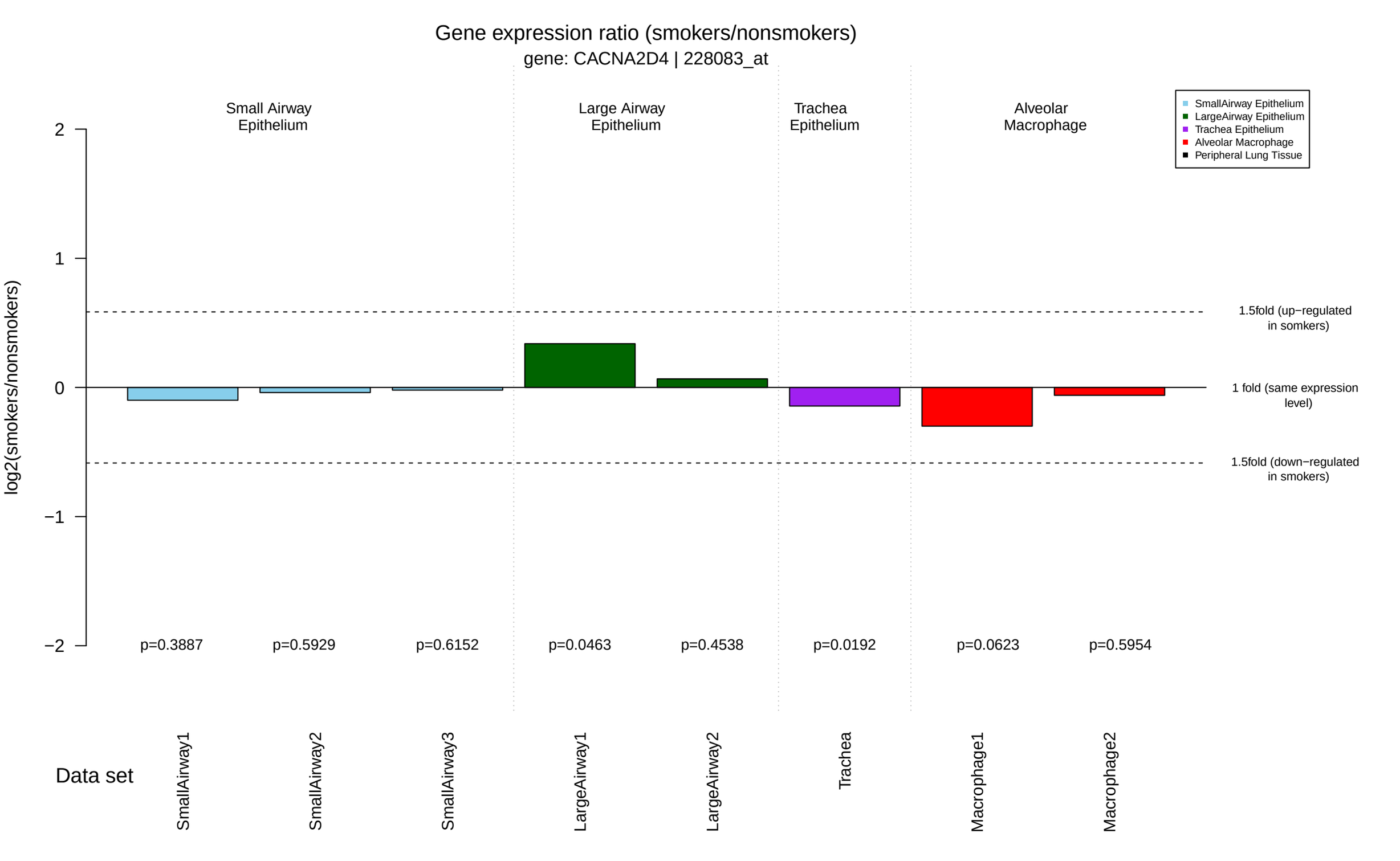
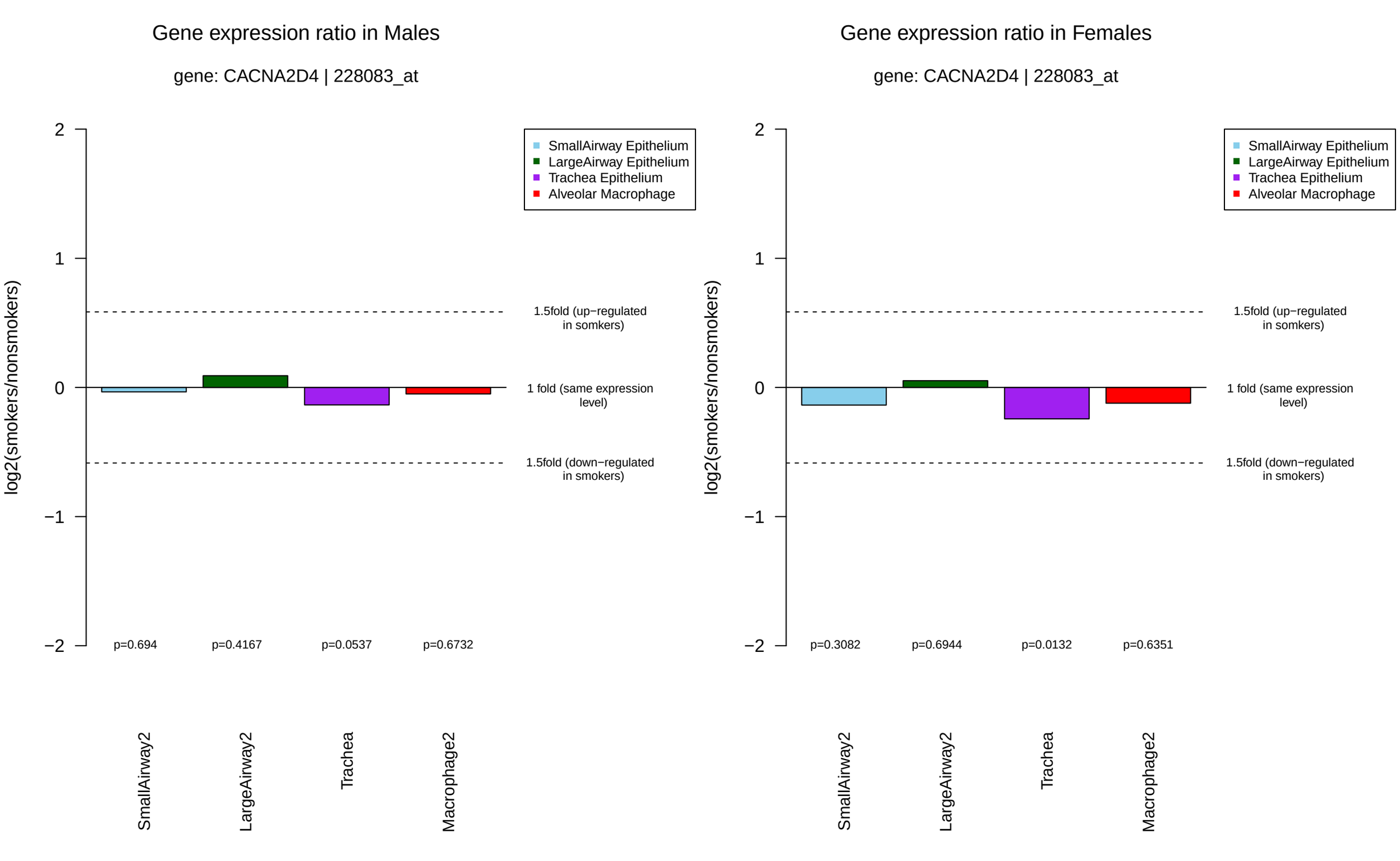
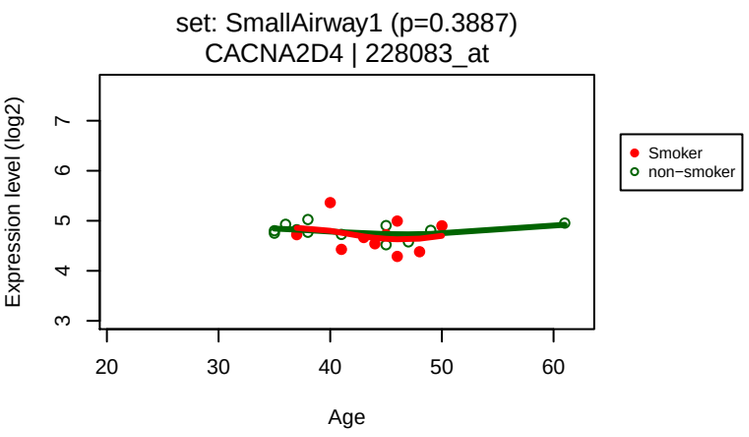
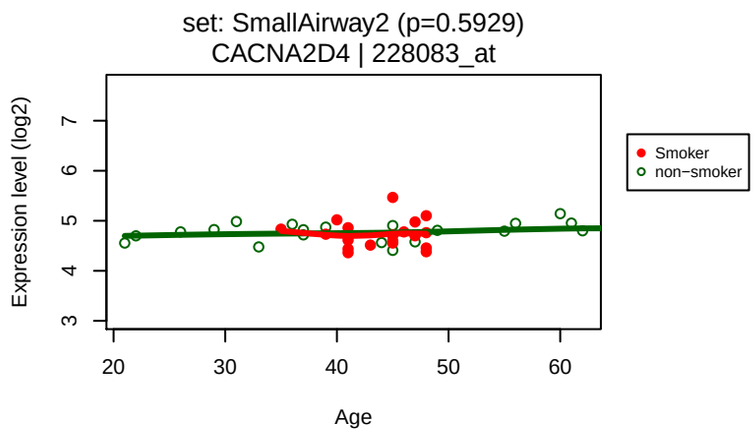
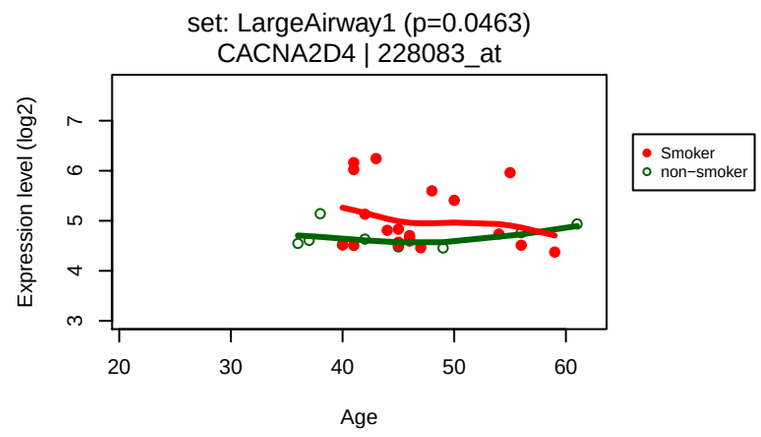

Gene expression of smokers and nonsmokers
gene: CACNA2D4 | 228083_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3887
p=0.5929
p=0.6152
p=0.0463
p=0.4538
p=0.0192
p=0.0623
p=0.5954
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

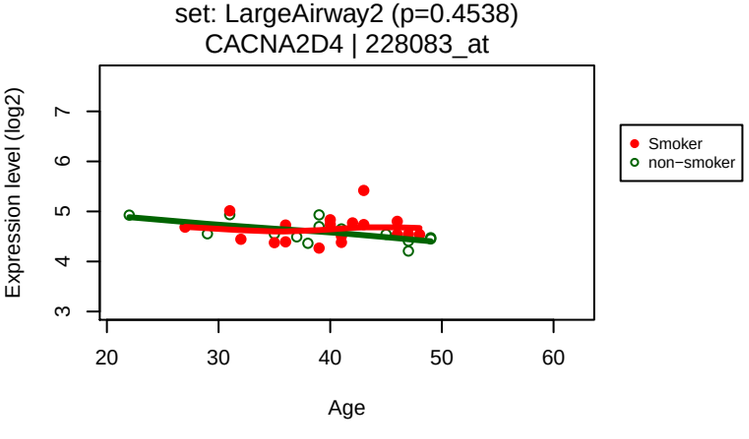

Gene expression ratio (smokers/nonsmokers)
gene: CACNA2D4 | 228083_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3887
p=0.5929
p=0.6152
p=0.0463
p=0.4538
p=0.0192
p=0.0623
p=0.5954
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

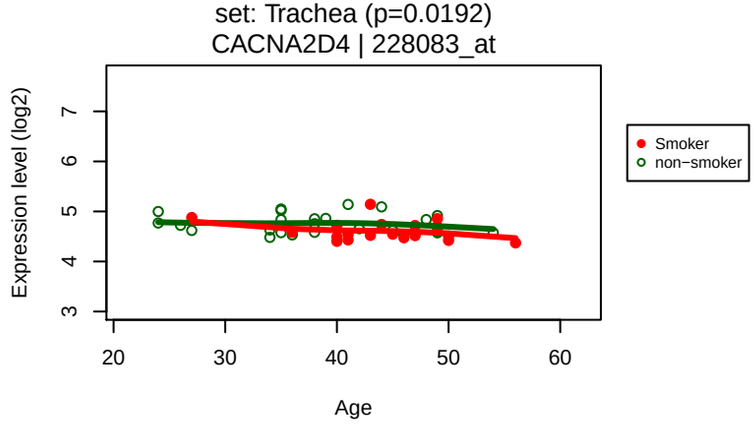

Gene expression ratio in Males
gene: CACNA2D4 | 228083_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.694
p=0.4167
p=0.0537
p=0.6732
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CACNA2D4 | 228083_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3082
p=0.6944
p=0.0132
p=0.6351
SmallAirway2
LargeAirway2
Trachea
Macrophage2

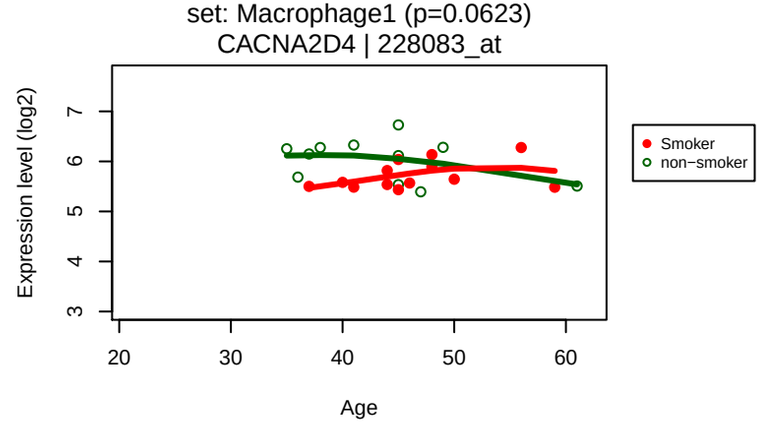

set: SmallAirway1 (p=0.3887)
CACNA2D4 | 228083_at
Expression level (log2)
Age
Smoker
non−smoker

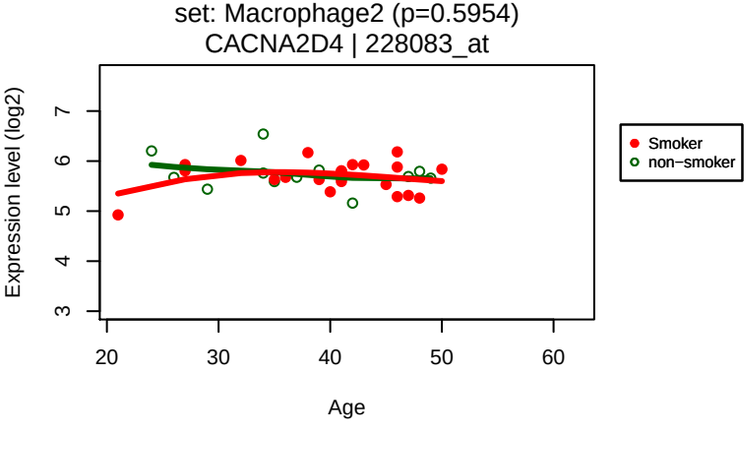

set: SmallAirway2 (p=0.5929)
CACNA2D4 | 228083_at
Expression level (log2)
Age
Smoker
non−smoker

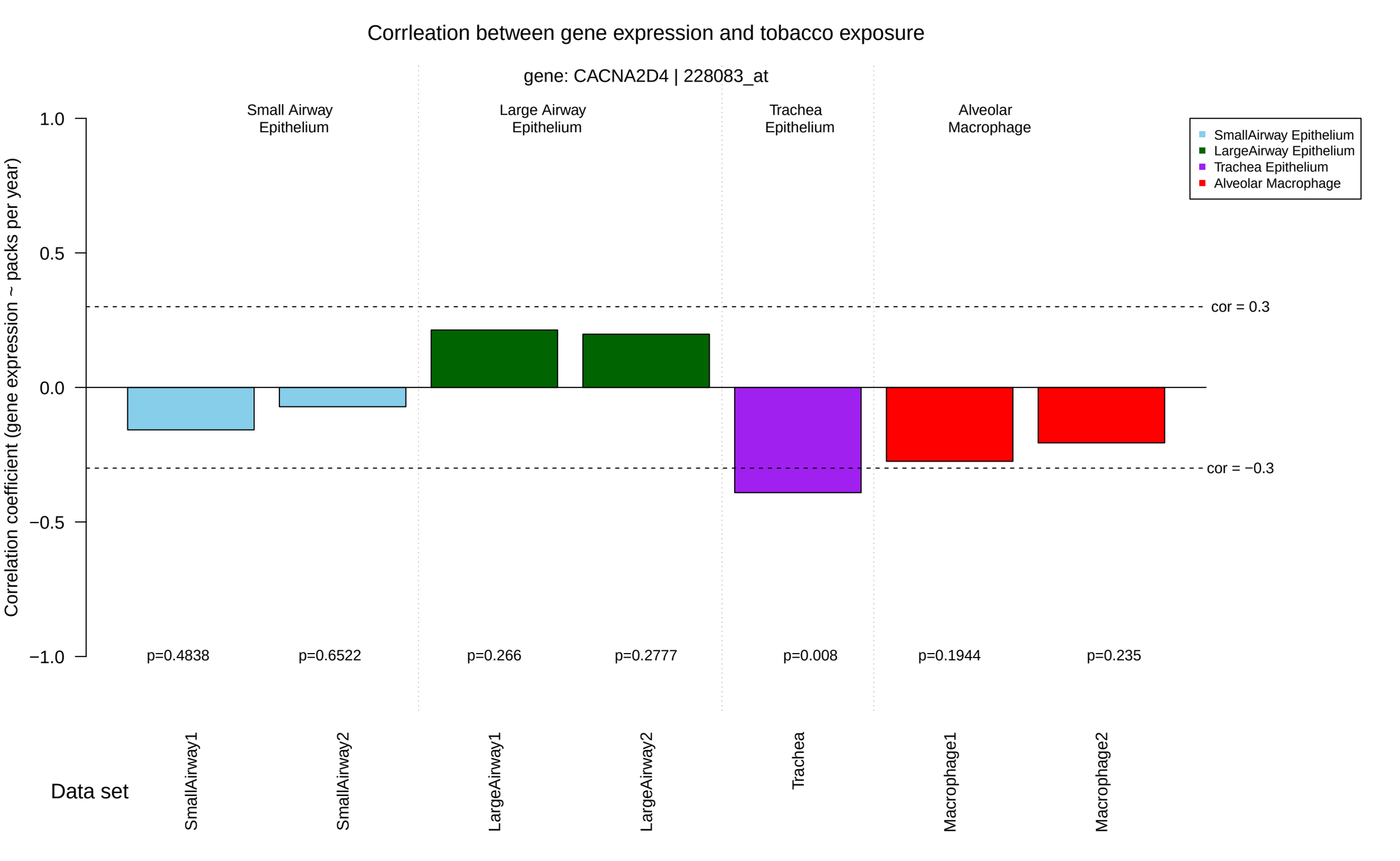

set: LargeAirway1 (p=0.0463)
CACNA2D4 | 228083_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4538)
CACNA2D4 | 228083_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0192)
CACNA2D4 | 228083_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0623)
CACNA2D4 | 228083_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5954)
CACNA2D4 | 228083_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CACNA2D4 | 228083_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4838
p=0.6522
p=0.266
p=0.2777
p=0.008
p=0.1944
p=0.235
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2